Sleep, Circadian Rhythms, and Trauma

Chapter 14

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Central question. Why is sleep so often disrupted after trauma, and why does treating sleep matter for recovery?

Learning objectives

- describe normal sleep architecture and regulation;
- explain the relationship between sleep and emotional memory;
- identify the main trauma-related sleep disorders;
- evaluate treatments for insomnia and nightmares;
- explain circadian disruption in displaced and shift-working populations.

Introduction

Sleep is not a passive state. It is an active process that consolidates memory, regulates emotion, restores immune and metabolic function, and clears metabolic waste. Trauma disrupts sleep, and disrupted sleep maintains trauma symptoms. This reciprocal relationship makes sleep both a consequence and a driver of illness, and it makes sleep an unusually tractable treatment target.

This chapter describes normal sleep, the effects of trauma on sleep, and evidence-based interventions. It also considers circadian rhythms, which are often overlooked in trauma care but are central to sleep, mood and metabolism.

> **A note on sleep measurement.** *Polysomnography measures brain activity, eye movements and muscle tone, and is the reference standard for sleep staging. Actigraphy estimates sleep from movement and is useful for patterns over time. Self-report is valuable but correlates imperfectly with objective measures. Research findings depend heavily on which method was used.*

14.1 Sleep architecture

Sleep consists of non-rapid eye movement (NREM) and rapid eye movement (REM) stages. NREM includes N1, N2 and N3 (slow-wave sleep). Sleep cycles through these stages roughly every 90 minutes, with more slow-wave sleep early in the night and more REM later.

Slow-wave sleep is associated with memory consolidation, particularly for declarative memory, and with growth hormone release and metabolic restoration. REM sleep is associated with emotional

processing, dream content and memory integration, although its functions remain debated.

Sleep architecture changes across the lifespan: slow-wave sleep declines with age, and sleep becomes more fragmented. These changes affect the interpretation of studies that compare trauma survivors with controls of different ages.

Sleep disturbance is now regarded as a central rather than secondary feature of PTSD, and may be a modifiable risk factor for poor outcome (Germain, 2013).

14.2 Two-process model

The two-process model describes sleep regulation by a homeostatic process and a circadian process. The homeostatic process tracks sleep pressure, which increases with time awake and dissipates during sleep. The circadian process is driven by the suprachiasmatic nucleus and regulates the timing of sleep and wakefulness.

The interaction of these processes determines sleep timing and quality. Irregular schedules, daytime napping and light exposure at night disrupt the interaction, producing insomnia-like symptoms even in people without trauma.

For trauma survivors, regularity is therapeutic. Consistent wake times, morning light and limited naps support the circadian system and reduce sleep pressure dysregulation.

14.3 Sleep and emotional memory

Sleep supports the consolidation of emotional memory and the integration of emotional experience. Some studies suggest that REM sleep specifically supports the processing of emotional content, reducing the affective charge of memories while preserving their informational content.

Sleep deprivation increases amygdala reactivity and reduces prefrontal control, producing emotional dysregulation. This is one mechanism by which insomnia maintains trauma symptoms: poor sleep makes emotional responses more intense and regulation more difficult.

The relationship is bidirectional. Trauma disrupts sleep, and poor sleep increases reactivity to trauma reminders. Treating sleep can therefore produce benefits beyond sleep itself.

14.4 Trauma and insomnia

Insomnia is common after trauma and is a risk factor for the development of PTSD. It includes difficulty initiating sleep, maintaining sleep and early waking, with associated daytime impairment.

Cognitive behavioural therapy for insomnia (CBT-I) is first-line treatment. It includes stimulus control, sleep restriction, cognitive restructuring and relaxation. It is effective in trauma populations, though clinicians sometimes worry that sleep restriction will exacerbate distress; with appropriate monitoring, it is usually well tolerated.

Medication can be useful short-term but carries risks of dependence, tolerance and falls, particularly in older adults. Antidepressants with sedating effects are sometimes used, but the evidence in PTSD is limited.

Insomnia frequently precedes and predicts PTSD, and targeted sleep treatment may accelerate recovery (Germain, 2013).

14.5 Nightmares

Nightmares are vivid, distressing dreams that awaken the sleeper and are remembered. They are common after trauma and can be chronic.

Imagery rehearsal therapy is a well-supported treatment. It involves recording the nightmare, changing its content to a less distressing version, and rehearsing the new version during the day. It reduces nightmare frequency and severity in many patients.

Other approaches include exposure, relaxation and medication. Prazosin, an alpha-1 antagonist, has been studied for trauma nightmares with mixed results; a large randomised trial in military veterans did not find a significant benefit over placebo (Raskind et al., 2018). This is an example of a plausible mechanism not translating into reliable clinical effect.

14.6 Sleep and fear extinction

Sleep supports extinction learning. In experimental studies, sleep after extinction training is associated with better retention of extinction memory, and REM sleep has been implicated.

This suggests that sleep interventions could enhance exposure therapy. Preliminary studies have tested this, but the evidence is not yet sufficient for routine clinical protocols. The practical implication is

simpler: protect sleep during trauma treatment, and avoid scheduling intensive sessions when the patient is severely sleep-deprived.

14.7 Circadian disruption

Circadian rhythms are synchronised by light, activity, meals and social cues. Disruption occurs with shift work, jet lag, hospitalisation, displacement and irregular routines.

Refugees and displaced people often experience multiple forms of circadian disruption: travel across time zones, irregular shelter conditions, night-time danger and loss of routine. These disruptions affect sleep, mood, metabolism and immune function.

Assessment should include sleep timing, light exposure, work schedules and living conditions. Interventions should be practical: morning light, regular meals, safe sleeping arrangements and reduced night-time light where possible.

14.8 Sleep in special populations

Sleep problems are common in children, older adults, pregnant people and people with comorbid medical conditions. Each group requires adaptation.

Children's sleep is affected by developmental stage, caregiving and environment. Nightmares and bedtime resistance are common after trauma, and parental involvement is essential. Older adults have more fragmented sleep and more medical comorbidity, and medication risks are higher. Pregnancy involves physiological sleep changes and specific safety considerations.

Institutional settings—shelters, hospitals, detention centres—often have conditions that make sleep difficult: noise, light, fear and lack of control. These are environmental problems that require environmental solutions.

14.9 Assessment of sleep

Assessment should include sleep history, sleep diary, screening for insomnia, nightmares, restless legs, sleep apnoea and circadian rhythm disorders. Comorbid conditions such as depression, anxiety, pain and substance use should be assessed.

Actigraphy can be useful for patterns over days or weeks. Polysomnography is indicated when sleep apnoea, narcolepsy or other primary sleep disorders are suspected.

Assessment should be trauma-informed: questions about sleep can evoke distressing content, and the bedroom may be associated with danger.

14.10 Sleep interventions

CBT-I is the first-line intervention for chronic insomnia. It is effective, durable and can be delivered in person, in groups or digitally. Imagery rehearsal therapy is first-line for nightmares.

Adjunctive strategies include sleep hygiene education, relaxation training and light therapy for circadian disturbance. Medication should be time-limited and reviewed regularly.

Sleep interventions should be integrated into trauma treatment rather than offered as an afterthought. Improving sleep can reduce reactivity, improve engagement and make trauma-focused work more tolerable.

14.11 Measuring sleep in clinical and research settings

Polysomnography records electroencephalography, eye movements, muscle tone, respiration, oxygen saturation and often cardiac activity. It is the reference standard for staging sleep and for diagnosing sleep apnoea, narcolepsy and periodic limb movement disorder. It is also expensive, unfamiliar and usually conducted for one night in a laboratory, which may not represent habitual sleep.

Home sleep apnoea testing is simpler and adequate for many cases of suspected obstructive sleep apnoea, but it underestimates severity when total sleep time is unknown. Actigraphy, usually a wrist device, estimates sleep and wakefulness from movement over days or weeks. It is useful for circadian patterns and for tracking change, but it misclassifies quiet wakefulness as sleep.

Self-report instruments, including sleep diaries and validated questionnaires such as the Insomnia Severity Index, capture the person's experience and its daytime consequences. They correlate imperfectly with objective measures, and that discrepancy is itself informative: sleep

state misperception is common in insomnia and should be addressed in treatment rather than dismissed.

Consumer wearables increasingly claim to measure sleep stages. Validation studies show wide variation between devices and settings, and proprietary algorithms are often undisclosed. They can support behavioural change, but they should not be treated as diagnostic. Preoccupation with device-derived sleep scores has been described as orthosomnia and can itself worsen insomnia (Baron et al., 2017).

14.12 Nightmares, dreams and emotional processing

Nightmares are vivid, disturbing dreams that awaken the sleeper and are remembered on waking. They are common after trauma, but not all post-traumatic nightmares replay the event; many involve themes of threat, loss or helplessness. This matters clinically, because imagery rehearsal therapy works with the nightmare as it occurs rather than requiring fidelity to the original event.

Why we dream remains unsettled. Threat-simulation accounts propose that dreams rehearse responses to danger; emotional-processing accounts propose that REM sleep supports the integration of affective experience; other accounts emphasise memory consolidation, network stabilisation or the random activation of circuits. None is established, and clinicians should avoid presenting any of them as fact.

What is better established is that nightmares can be treated. Imagery rehearsal therapy reduces nightmare frequency and distress, and it can be delivered in groups or digitally. Treating nightmares often improves sleep quality and daytime functioning, and it may improve engagement with trauma-focused therapy.

Dream interpretation as a diagnostic tool is not supported. Dreams can be meaningful to the dreamer, and exploring them can be therapeutic, but they do not provide reliable access to hidden memories or to a definitive account of the past.

Synthesis

Table 14.1 summarises sleep problems and interventions.

Table 14.1. Sleep problems and evidence-based responses

Problem	Features	First-line intervention	Notes
Insomnia	Difficulty initiating/maintaining sleep	CBT-I	Effective in trauma populations
Nightmares	Vivid awakening dreams	Imagery rehearsal therapy	Prazosin evidence mixed
Circadian disruption	Irregular timing	Light, routine, schedule	Consider social context
Sleep apnoea	Breathing pauses	CPAP, specialist referral	Often missed
Restless legs	Urge to move at night	Iron studies, specialist care	Comorbid with depression

Composite case study

A patient reports sleeping four hours a night, waking with nightmares and fearing sleep. Treatment begins with CBT-I and imagery rehearsal therapy, alongside trauma-focused therapy. Sleep improves over eight weeks, and the patient reports better tolerance of therapy sessions. The clinician notes that improved sleep is not the same as recovery, but it makes recovery more feasible.

Review and discussion questions

How do the homeostatic and circadian processes interact to regulate sleep?

Why is insomnia a risk factor for PTSD rather than only a consequence?

What is the evidence for imagery rehearsal therapy?

Why did prazosin trials produce disappointing results?

How should sleep interventions be adapted for refugees and displaced populations?

What are the risks of sedative medication in older trauma survivors?

Research assignments

- Design a trial of CBT-I delivered before trauma-focused therapy.
- Critically appraise the evidence on sleep and extinction consolidation.
- Write a sleep hygiene guide that is accurate, practical and trauma-informed.

Chapter summary

Sleep is an active process essential for memory, emotion regulation, immunity and metabolism. Trauma disrupts sleep, and poor sleep maintains trauma symptoms. Insomnia and nightmares are treatable with CBT-I and imagery rehearsal therapy. Circadian disruption is common in displaced and shift-working populations and requires practical intervention. Sleep should be assessed and treated as part of trauma care, not as an optional extra.

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